

CITS3005 Knowledge Representation Homework 2

September 9, 2026

Overview

This homework assignment requires you to build and reason on a knowledge graph. You will need to use `rdflib` and `problog`.

- **Worth:** 10% of final grade
- **Due:** 11.59pm, Wednesday September 30, 2026.
- **Submission:** via LMS, see the submission page for instructions
- **Demonstration to teaching staff:** Workshop on October the 8th.

To receive a mark for this homework you must both submit your work by the deadline and attend your allocated Workshop time in Week 11 (October 8) to answer questions about your work. If you cannot attend your workshop due to unavoidable commitments you must let the unit coordinator know (email: cits3005-csse@uwa.edu.au) before the deadline in order to make alternate arrangements.

See LMS for further guidance re submission, the late policy, and GenAI policy.

Tasks

This homework exercise requires you to build your own knowledge graph about a fictional world based on a tv show, movie, or book of your choosing. You can think of it as the knowledge graph backing (part of) a wiki about your favourite fictional world! You should choose something that you have a good understanding, and something that is not too complex (e.g. you may choose to focus on a particular subset of characters/places/events etc):

- Ideally there should be between 10 and 100 entities (but there can be more literals)
- There should be between three and 10 types of entities.
- There should be at least three different types of relations.

Part I: Descriptions

Begin by completing the following tasks to establish what you are trying to build:

1. Describe the world for which you are building a knowledge graph. Give a textual description of each of the types and each of the relations that your knowledge graph will use. 2 marks
2. Describe a set of useful queries that your knowledge graph could be used to answer. 1 mark

You should ensure the types and relations you decide on for (1) are enough to reason on some interesting queries in (2). This should be submitted as a pdf report (less than 1000 words).

Part II: RDF and SPARQL

Next, implement your knowledge graph and queries using RDF and SPARQL

3. Construct the knowledge graph using `RDFLib`. 2 marks
4. Demonstrate your queries using `SPARQL`. 2 marks

This should be submitted as an RDF/XML file (the final knowledge graph) and `.py` file (any code used to build the knowledge graph and execute the queries)

Part III: Probabilistic Reasoning

Say you want to extend your wiki to include a speculation section about an uncertain past or future event or outcome (e.g. the likelihood a character will share a secret, a treasure is in a certain place, characters previously interacted/meet again/become friends or allies etc). As a final task:

5. Use problog to model the uncertain event/outcome. Your program must: 3 marks

- Use entities and relations from your knowledge graph (represented in problog). You do not need to reimplement your entire knowledge graph from Part I and II, just a subset that is useful for any probabilistic reasoning you need to do.
- Include at least two new first-order probabilistic clauses.
- Contain at least two sample probabilistic queries.
- Make clear in comments what uncertain event you are modelling and any assumed probabilities for clauses/facts.
- You should not turn any of your known facts from Part I or II into probabilistic facts/clauses, as this would be modelling uncertainty when there is none!

As a guide, the social networks problem in the problog tutorial is a good example of the minimum level of complexity for the probabilistic clauses/facts for full marks. This should be submitted as a .pl file.